How Economists Discuss Central Bank Digital Currencies

Themes, Tone and Stance, 2014–2026

Louis Drogo Kenza Redwane Adriel Mazere

M2 – Private crypto
October 16, 2026

Research question

"How has economists and institutional discourse on CBDCs evolved between 2014 and 2026, and to what extent does this evolution align with changes in the policy environment?"

Abstract

Keywords:
Number of pages: 16.2 (main text, Introduction to Conclusion, figures excluded)
Appendix: 10 pages

Contents

1 Introduction **1**

2 Related literature **1**
- 2.1 Central bank digital currencies . . . 1
- 2.2 What the field builds on . . . 1
- 2.3 Bibliometric and text-as-data studies of CBDCs . . . 2

3 Data **2**
- 3.1 Sources and collection . . . 2
- 3.2 De-duplication and enrichment . . . 3
- 3.3 Relevance: is CBDC the theme? . . . 3
- 3.4 Who counts as an economist . . . 3
- 3.5 Descriptive statistics . . . 4

4 Who writes **5**
- 4.1 Method: co-citation, co-authorship, bibliometric laws and distinctive vocabulary . . . 5
- 4.2 Findings . . . 6

5 What is written **7**
- 5.1 Method: thematic map, topic model and covariate-adjusted shares . . . 7
- 5.2 Findings . . . 9

6 How economists discuss CBDCs: tone and stance **11**
- 6.1 Method: tone (FinBERT) and stance (reading and a local language model) . . . 11
- 6.2 Findings . . . 12
- 6.3 Do policy events move the tone? . . . 12

7 Why: stated motivations **13**
- 7.1 Method: a validated codebook . . . 13
- 7.2 Findings . . . 13

8 When: the policy environment **14**
- 8.1 Method: bursts, event study with placebo dates, difference-in-differences . . . 14
- 8.2 Findings . . . 15

9 The discussion and its environment **15**
- 9.1 Period by period . . . 15
- 9.2 What can and cannot be said about causes . . . 16

10 Robustness **16**

11 Discussion **17**
- 11.1 Limitations . . . 17

12 Conclusion **18**

References **19**

A Search vocabulary **A-1**

B Harvesting algorithms **A-1**
B.1 Sources . . . A-1
B.2 Pagination and rate limits . . . A-2
B.3 Press: recursive bisection of time windows . . . A-2
B.4 Full text, enrichment and de-duplication . . . A-3
B.5 Relevance score . . . A-3
B.6 Who counts as an economist . . . A-4

C Validation samples and coding guide **A-4**
C.1 Coding guide . . . A-4
C.2 Results . . . A-5

D Additional tables and figures **A-7**

E Data and code availability **A-9**
E.1 Data . . . A-9
E.2 Code . . . A-9
E.3 Software and models . . . A-9

F Use of artificial intelligence and statement of originality **A-10**
F.1 Use of artificial intelligence . . . A-10
F.2 Statement of originality . . . A-10

1 Introduction

2 Related literature

2.1 Central bank digital currencies

A central bank digital currency (CBDC) is a digital liability of a central bank that can be used as a means of payment and store of value. Although the term CBDC is recent, the idea of giving the public access to central-bank money in a non-cash form is older. For instance, Tobin (1987) proposed "deposited currency accounts", which would combine the convenience of bank deposits with the safety of central-bank money.

The modern CBDC literature developed much later, particularly as digital payments expanded and central banks began considering their own digital currencies. Early contributions focused mainly on the implications of CBDCs for monetary policy, banking and financial stability. Barrdear and Kumhof (2016) study the macroeconomic consequences of issuing an interest-bearing CBDC, while Bordo and Levin (2017) discuss its potential role in the implementation of monetary policy. A parallel literature examines the possible effects on commercial banks, in particular the risk that households substitute CBDCs for bank deposits, with consequences for bank funding and financial intermediation.

These questions progressively led to a broader literature on how CBDCs should be designed. The design of a CBDC involves several choices, including whether it should be interest-bearing, account- or token-based, widely accessible, and how privacy should be protected. These choices can create trade-offs between payment efficiency, financial inclusion and the risks of bank disintermediation or financial instability. Agur, Ari and Dell'Ariccia (2019), for example, study these trade-offs, while Bindseil (2020) discusses how a tiered remuneration system could limit large transfers from bank deposits into CBDCs

More recently, the debate has expanded beyond monetary policy and banking. As central banks started experimenting with concrete projects, questions related to payment efficiency, financial inclusion, privacy, cross-border payments and monetary sovereignty became increasingly important. The literature therefore moved from asking mainly what a CBDC could do to the monetary and banking system toward a broader discussion of how it should be designed and what policy objectives it should serve.

Our paper takes a different perspective from this literature. Rather than estimating one particular economic effect of CBDCs, we study how the economic and institutional discussion surrounding them has itself evolved over time.

2.2 What the field builds on

The CBDC literature can be organised around three recurring tensions that predate CBDCs themselves.

The first is the tension between privacy and state-issued money. Chaum (1983) showed how cryptographic techniques could enable electronic payments while preserving users' privacy, while Nakamoto (2008) later demonstrated that digital payments could operate without a trusted central intermediary. CBDCs bring these questions into a different institutional setting: can a central bank provide digital money while preserving the degree of privacy traditionally associated with cash?

The second is the tension between financial efficiency and financial stability. The banking literature has long studied the role of banks in providing liquidity and transforming deposits into credit (Diamond and Dybvig, 1983), while proposals such as Tobin's (1987) deposited urrency accounts questioned whether the public should have access to a safer form of central-bank money. BDCs revive this trade-off. A widely accessible central-bank liability may improve the safety and efficiency of payments, but it may also substitute for commercial-bank deposits and affect bank intermediation and credit provision

(Andolfatto, 2020; Brunnermeier and Niepelt, 2019).

The third concerns monetary sovereignty in an increasingly digital and international monetary system. The development of cryptocurrencies, stablecoins and global private payment initiatives has raised questions about the coexistence of public and private forms of digital money and about the ability of central banks to preserve monetary sovereignty. These concerns have increasingly extended the CBDC debate beyond monetary policy and payment design towards cross-border payments and geopolitical considerations.

These tensions are not separate strands of the literature, but overlapping questions that connect monetary economics, banking and digital payments. Our paper does not attempt to resolve them. Instead, it studies how the relative importance of these questions in economists' discourse has evolved between 2014 and 2026.

2.3 Bibliometric and text-as-data studies of CBDCs

Within our corpus, we observe 29 titles that announce a literature, systematic or bibliometric review of CBDCs, published between 2019 and 2026. When comparing the five most recent bibliometric studies with ours, we found three shared main limits. First, their data come from one or two databases, Scopus and/or Web of Science, and are mostly journal papers written in English. Working papers, central-bank speeches and central-bank documents are therefore mostly absent from their analysis, so that their sample is very limited compared to ours: a few hundred documents compared to 4,710. In addition, they build their analysis with standard tools that mainly describe what is studied, but not how (tone and stance) or why. Finally, their classifications are never checked against a reading of the texts themselves.

Our study addresses these three limits. Our corpus is much larger, with papers coming from five databases, central-bank speeches and documents, in different languages, and every text is checked to be written by an economist or an economic institution. We ask not only what is written but also who writes it, how, why and when. Finally, each automatic classification we use is compared with our own reading of a sample of texts.

MISSING : text-as-data studies

3 Data

3.1 Sources and collection

In order to properly use the tools and for better selection purposes, we try first to fetch as many documents as possible. Indeed, if we had selected the sample based on pre-determined criteria of our choice, we could face the bias of pre-selecting the sample. By having a larger sample with partly irrelevant data, we can apply selection processes based on quality criteria, such as the relevance of a document. Also, it is useful to have a large sample to be able to validate results on sub-samples or parts of the data kept for testing, which strengthen the results, or for robustness tests.

When building the fetching algorithm, we therefore tried to reach for selection criteria as broad as possible. First, we wanted many "types" of documents, namely: official academic papers published in journals or magazines, press articles, speeches from institutional figures or official documents. The sources span :

- Academic databases: OpenAlex, Semantic Scholar, arXiv, HAL and DOAJ, which mainly contain working papers and journal articles, but also books.
- Institutional databases: speeches were mainly extracted from the BIS archive (as well as BIS RSS)

and ECB's speech database. We also extracted from 26 central banks', the EU institutions and international institutions sites the pages relating to CBDC.

- Press: press articles are extracted from Google News

The searching process is fairly easy: we choose 115 sentences in 12 langues (using LLM and suggestions, as well as our own formulations, based on common expressions in the literature), that are used to search accross titles and abstracts alike. The APIs return results page by page for the papers and authorities' websites and the press articles are harvested using a recursive bisection to single down the relevant articles. Everything from all sources is collected and we can go over the filtering process.

3.2 De-duplication and enrichment

Because some articles may be duplicates or pre-publications or have similar names, we try to get rid of all duplicates or fuse when they are from the same authors and relate to the same paper. The contents of the documents are also enriched, in that we also harvest the language in which it is procuded, the keywords of the text (using YAKE), the exact date (when possible, if not indexed at the beginning of the year issued), the authors' names and media in which it has been issued, the introduction and conclusion for the papers, which resume the motivations and results of the paper, as well as the content of the text.

3.3 Relevance: is CBDC the theme?

Because CBDC is also a skin disease (Chronic Bullous Disease of Childhood) and a chemical term (CycloButane DiCarboxilate), or that it can be used as an acronym for anything that the author does not explicit in the titles and abstracts, it is possible to obtain documents unrelated to the monetary concept. Hence, using a range of contextual vocabulary (such as currency, payment, central bank, monnaie, Geld, etc.), we filter for papers in which the term CBDC is ambiguous and not directly related to Central Bank Digital Currency.

More importantly, we want CBDC to be the main theme of the paper, and not merely a cited reference related to the actual content. Therefore, we establish relevance scoring as such: each field $f \in F$ gets centrality score $c_f \in [0, 1]$ (this term will be also used later in the paper, but with a different meaning, be cautious !) that gives weight to the paper relevance if CBDC is a central theme to the paper. The centrality scores are determined as such:

$$c_{title} = \mathbf{1}[\text{CBDC in title}] \qquad c_{kw} = 0.4 \cdot \mathbf{1}[\text{CBDC keyword}]$$
$$c_{abs} = \frac{\text{card(sentences mentioning CBDC)}}{\text{card(sentences)}} \qquad c_{body} = 0.5 \min\left(1, \frac{\rho}{3}\right) + 0.5\sigma$$

Where ρ = mentions per 1000 words and σ = share of ten slices with a mention. And the relevance score is:

$$\text{score} = 100 \cdot \left(0.6 \max_{f \in F} c_f + \frac{0.4}{|F|} \sum_{f \in F} c_f \right) \tag{1}$$

3.4 Who counts as an economist

We want now to ensure that the text has been written by an economist. This is fairly easy since we can just check using the authors' ID for each academic sources, which tells whether the author is an economist or not. Therefore, all research document have been kept, when the relevance score clearly

indicated that the theme was CBDC. Moreover, since articles and speeches issued by authorities are produced by either economists working there or economists by practice (C. Lagarde is a lawyer for example, but her vast experience as head of IMF and of ECB qualifies her as such). Press articles and documents produces by governments were therefore dropped.

3.5 Descriptive statistics

The corpus analysed in this paper holds the 4,710 texts with CBDC as a theme that were written by economists (Table 1). Research makes up 85% of it (3,994 texts); central bank speeches (511) and documents of central banks and international institutions (205) make up the rest. For three texts in four, CBDC is the main theme.

The first texts by economists date from 2016, and the field took off after 2019 (Figure 1). Research grew from 20 texts in 2017 to 839 in 2025, a compound rate of about 60% a year, and 2026 had already exceeded 2025 by October. Speeches rose until 2022 (77) and have stayed close to 70 a year since. Documents of institutions jumped in 2023 (49), the year of the European Commission's proposal for a digital euro and of the end of the ECB's investigation phase. The first period of the paper, 2014–2019 rests on a small number of texts (211 texts, against 972 in 2020–2022 and 3,360 in 2023–2026), which is why changes are estimated with intervals and tested against random dates (Sections 6 and 8).

The growth rates above are compound annual rates: with N_y texts in year y, the rate between y_0 and y_1 is

$$g = \left(\frac{N_{y_1}}{N_{y_0}}\right)^{1/(y_1 - y_0)} - 1, \tag{2}$$

so that research, with $N_{2017} = 20$ and $N_{2025} = 839$, grew by $g = (839/20)^{1/8} - 1 \approx 60\%$ a year.

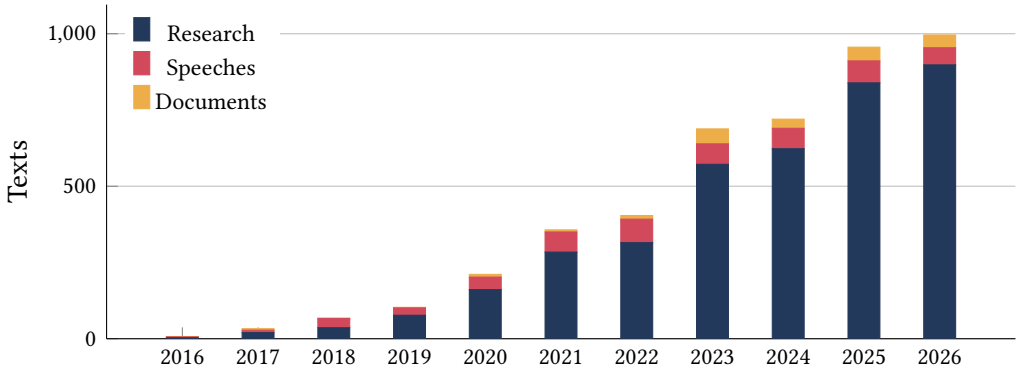


Figure 1. Texts written by economists, by year and type (2026: until October)

Research is spread over 1,479 outlets, and working papers weigh as much as journals: the largest outlets are SSRN (461 texts) and RePEc (164), ahead of any journal. Articles account for 2,455 research texts, preprints for 583, book chapters for 394. Citations are highly concentrated: the median research text has not yet been cited, while the mean is 5.5 citations, so a small core of papers receives most of the attention (Section 4). On the institutional side, 61 institutions are represented, led by the European Central Bank (216 texts), the Deutsche Bundesbank (84), the Banque de France (49), the Bank of Japan (41) and the Sveriges Riksbank (32): the digital euro makes the euro area the centre of the institutional debate.

Texts are written in 39 languages. English dominates (79%), but Russian is a substantial second language (422 texts, 9%), mostly on the digital ruble, ahead of German, Ukrainian and Spanish. English-only databases miss about a fifth of the economists' writing on CBDCs.

Most texts can be dated to the day (71%), and most have keywords (85%). An abstract is available for 68% of texts, an introduction for 62% and the full text for 63%, while reference lists, needed for co-citation, exist for 37% of research texts only. Each analysis uses the texts for which its input exists, and the number of texts is reported with every result.

Table 1. The economists' corpus

	Texts		**Texts**
Type of text		*Language (39 in all)*	
Research	3,994	English	3,721
Central bank speeches	511	Russian	422
Documents of institutions	205	German	94
Total	4,710	Ukrainian	91
Relevance		Spanish	74
CBDC the main theme	3,513	French	43
CBDC a secondary theme	1,197	*Fields available*	
Research, by document type		Exact day of publication	71%
Article	2,455	Abstract	68%
Preprint	583	Introduction	62%
Book chapter	394	Full text	63%
Conference paper	131	Keywords	85%
Dissertation	117	Reference list (research)	37%

4 Who writes

4.1 Method: co-citation, co-authorship, bibliometric laws and distinctive vocabulary

In the following analysis, to see who writes about CBDCs (which authors, institutions and papers), we use four different tools. The first one, called a co-citation network, links references that are cited in the same paper, which allows us to build groups of references often cited together. Then, we build a co-authorship network, which shows research teams by linking authors when they write a paper together. In both networks, to avoid giving too much weight to links between very popular references or authors, we weigh each link by its association strength, which takes into account how popular each item is.

For two items i and j (references or authors), c_{ij} is the number of papers in which they appear together and c_i the number of papers in which i appears. The weight of the link is the association strength given by:

$$s_{ij} = \frac{c_{ij}}{c_i\, c_j}, \tag{3}$$

proportional to the ratio of observed to expected co-occurrences (van Eck and Waltman, 2009). Groups are found with the Louvain algorithm, which maximises the modularity

$$Q = \frac{1}{2m} \sum_{i,j} \left[s_{ij} - \frac{k_i k_j}{2m} \right] \delta(g_i, g_j), \tag{4}$$

where $k_i = \sum_j s_{ij}$, $m = \frac{1}{2} \sum_{i,j} s_{ij}$, and $\delta(g_i, g_j) = 1$ when i and j are in the same group. This metric effectively evaluates the quality of the network's division into "communities".

Besides these two networks, we also compute two standard bibliometric laws. The first one, Lotka's law, allows us to see how papers are spread among authors, whether a few authors write most of the papers or whether many authors only write one. Then, the second one, called Bradford's law, ranks

publication sources by their number of papers on CBDCs and splits them into three groups of equal sizes in papers. This shows whether the literature is concentrated in a few sources or spread over many of them.

Lotka's law states that the share of authors with n papers decreases as a power of n,

$$f(n) = \frac{C}{n^{\beta}}, \tag{5}$$

where $f(n)$ is the share of authors who write exactly n papers on CBDCs, C a normalising constant and β the Lotka exponent, estimated by least squares on $\log f(n) = \log C - \beta \log n$; $\beta \approx 2$ in mature fields, and a larger β means that more authors write a single paper. For Bradford's law, sources are sorted by decreasing number of papers and cut into three zones holding a third of the papers each, and the number of sources in each zone, $n_1 : n_2 : n_3$, measures concentration.

Finally, in order to compare the words used by research economists and central bankers, we use a method called log-odds ratio with an informative Dirichlet prior. Compared with simple word counts, this method does not give too much weight to rare words and allows us to see which words one group uses more than the other.

Let y_{gw} be the number of times group g uses word w, n_g the total number of words of group g, and $\alpha_w = \alpha_0 \, \pi_w$ a prior proportional to the frequency π_w of w in the whole corpus. For group i against group j (Monroe, Colaresi and Quinn, 2008),

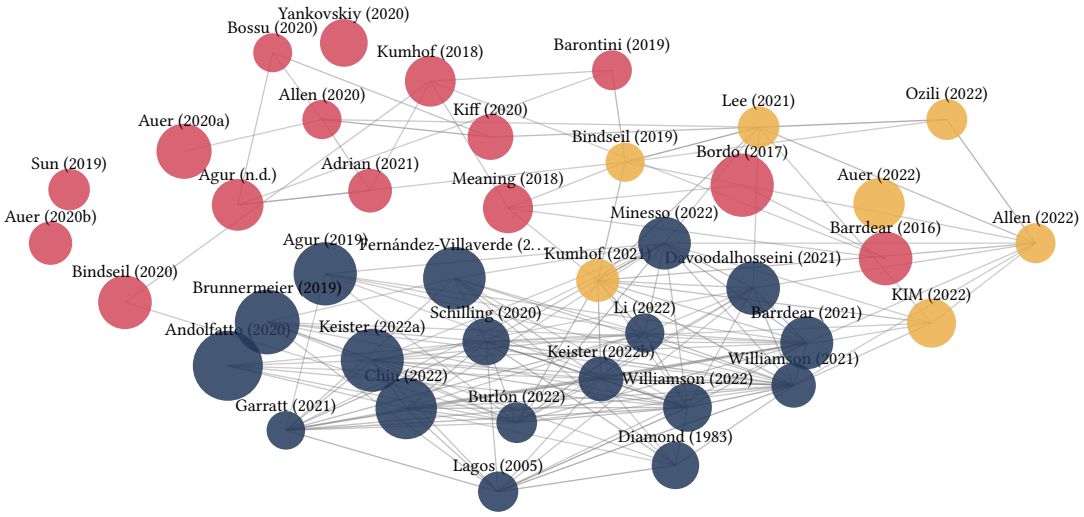
$$\delta_w = \log \frac{y_{iw} + \alpha_w}{n_i + \alpha_0 - y_{iw} - \alpha_w} - \log \frac{y_{jw} + \alpha_w}{n_j + \alpha_0 - y_{jw} - \alpha_w}, \qquad z_w = \frac{\delta_w}{\sqrt{\dfrac{1}{y_{iw} + \alpha_w} + \dfrac{1}{y_{jw} + \alpha_w}}}, \tag{6}$$

and words are ranked by z_w. The prior pulls the log-odds of rare words towards zero, so that a word used twice by one group and never by the other does not come first.

4.2 Findings

From our analysis, the two largest sources are SSRN (461 papers) and RePEc (164 papers), two repositories of working papers. We also obtain a Lotka exponent of 3.3, which is higher than the value, of about 2, found in more mature fields: 83% of authors write only one paper on CBDCs. In the same way, Bradford's law shows that only 21 sources publish a third of the papers, while the last third is spread over more than 1,100 sources. In short, a small group of authors and sources publishes regularly on CBDCs, while most economists only write on the subject once.

Papers with at least one author from a central bank, the BIS, the IMF or the World Bank represent 10% of the papers but receive 48% of the citations, and their share of the papers fell from 23% in 2014-2019 to 7% in 2023-2026. In other words, these economists write few papers but are cited almost five times more than their weight in the literature, while most new papers now come from economists outside these institutions. The co-citation network gives three groups of references, around Andolfatto (2020) and Brunnermeier (2019), Bordo (2017) and Barrdear (2016), and Auer (2022). However, these groups are not clearly separated since we have a low of $Q = 0.10$: the literature does not split into distinct schools of thought. In addition, the co-authorship network shows that research teams are small: about half of the papers are written by a single author, and only 15% involve authors from different countries. Among institutions, the European Central Bank is by far the most active, with 118 speeches and 98 official documents. On the research side, Russia is the most represented country: it publishes the most papers, and four of the eight most active research institutions are Russian.



Groups (largest members): • Andolfatto (2020), Brunnermeier (2019), Agur (2019), Keister (2022); • Bordo (2017), Auer (2020a), Barrdear (2016), Bindseil (2020); • Auer (2022), Kim (2022), Kumhof (2021), Lee (2021). 1,493 papers with reference lists, 13,810 distinct references; the 40 most co-cited references are shown; links weighted by association strength; Louvain clusters, $Q = 0.10$.

Figure 2. Co-citation network of research by economists

Finally, we observe that research economists and central bankers do not use the same words. Research economists use more words such as *legal*, *analysis*, *model* and *impact*, while central bankers use more words such as *payments*, *euro* and *Europe*. In other words, research economists seem to study CBDCs from a more analytical point of view, while central bankers mostly talk about payments and the digital euro.

Table 2. Words most over-used by each group

Research economists		Central bankers	
legal	23	payments	28
analysis	21	euro	22
development	21	european	18
research	20	europe	18
monetary	20	area	18
impact	19	services	15
model	18	innovation	14
implementation	17	conference	14

z-scores of the log-odds ratio with an informative Dirichlet prior, each group against the other. English texts only; research economists: title and abstract (259,105 words); central bankers: speeches (232,446 words). Words defining the topic (CBDC, digital, currency…) are removed.

5 What is written

5.1 Method: thematic map, topic model and covariate-adjusted shares

To see what is written by economists, we use two methods. The first one consists in grouping keywords that often appear together in the same papers into themes. Then, we place these themes on a thematic map with two axes: how much a theme is linked to the other themes (*centrality*) and how much its keywords are linked to each other (*density*). This allows us to classify themes into four types: motor, basic, niche, and emerging or declining.

Formally, with c_{ij} the number of papers where keywords i and j appear together and c_i the number

of papers with keyword i, the equivalence index is $e_{ij} = c_{ij}^2/(c_i c_j)$. For a theme C (a group of keywords found as in Equation (4)),

$$\text{centrality}(C) = 10 \sum_{i \in C,\ j \notin C} e_{ij}, \qquad \text{density}(C) = \frac{100}{|C|} \sum_{i,j \in C} e_{ij}, \tag{7}$$

where $|C|$ is the number of keywords in C, and the four quadrants are separated by the medians of the two measures (Callon, Courtial and Laville, 1991; Cobo et al., 2011).

The second method consists in grouping texts by their meaning. To do so, we use *SBERT*, a multilingual model that turns each title and abstract into a vector, so that two similar texts end up close, even if they are not written in the same language. We then group these vectors into 24 topics with *BERTopic* and label each topic with its most specific words.

Each text t is represented by a vector $\mathbf{e}_t \in \mathbb{R}^{384}$, and the similarity of two texts s and t is the cosine

$$\cos(\mathbf{e}_s, \mathbf{e}_t) = \frac{\mathbf{e}_s \cdot \mathbf{e}_t}{\|\mathbf{e}_s\| \, \|\mathbf{e}_t\|}. \tag{8}$$

The vectors are reduced to five dimensions with UMAP and grouped with HDBSCAN, which finds dense regions without fixing the number of topics in advance. Each topic c is then described by the words with the highest class-based TF-IDF,

$$W_{w,c} = \frac{\text{tf}_{w,c}}{|c|} \log\left(1 + \frac{A}{f_w}\right), \tag{9}$$

where $\text{tf}_{w,c}$ is the number of occurrences of word w in topic c (all texts of the topic put together), $|c|$ the total number of words in topic c, f_w the number of occurrences of w in all topics and A the average number of words per topic (Grootendorst, 2022).

Finally, to make sure that a change in the share of a topic is not only due to a change in who writes, we run a logit regression of each topic on the period, the type of author and the language. That allows us to measure how the share of each topic changed between two periods (here, between 2014-2019 and 2023-2026), in percentage points, all else being equal.

For text t and an outcome $y_t \in \{0, 1\}$ (here, belonging to a topic),

$$P(y_t = 1) = \Lambda\big(\alpha + \beta_{\text{period}(t)} + \gamma_{\text{actor}(t)} + \theta\, \mathbf{1}[\text{English}_t]\big), \qquad \Lambda(x) = \frac{1}{1 + \exp(-x)}, \tag{10}$$

where α is the intercept; β_p is the effect of period $p \in \{2020\text{–}2022, 2023\text{–}2026\}$, with 2014–2019 as the reference ($\beta_{2014\text{–}2019} = 0$); $\gamma_{\text{actor}(t)}$ is the effect of the type of author of t, equal to 0 for research economists and to γ for central banks and institutions; $\mathbf{1}[\text{English}_t]$ equals 1 when t is written in English and 0 otherwise, and θ is its effect; Λ is the logistic function. The model is estimated by maximum likelihood with robust (HC1) standard errors. The change reported is the average marginal effect of the period,

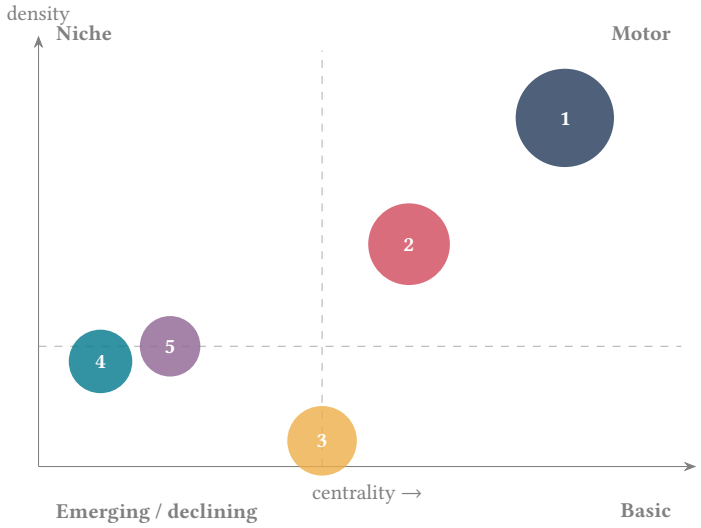
$$\Delta = \frac{1}{T} \sum_{t=1}^{T} \Big[\hat{P}\big(y_t = 1 \mid 2023\text{–}2026, x_t\big) - \hat{P}\big(y_t = 1 \mid 2014\text{–}2019, x_t\big)\Big], \tag{11}$$

where T is the number of texts, $\hat{P}(y_t = 1 \mid p, x_t)$ is the probability predicted by Equation (10) for text t when its period is set to p, and x_t holds the actor and the language of text t, kept at their observed values; its 95% confidence interval is obtained by drawing the coefficients 400 times from their estimated distribution. This is the prevalence part of a structural topic model (Roberts et al., 2014), applied to the

topics found by BERTopic.

5.2 Findings

The thematic map shows that the CBDC literature is driven by two themes: blockchain technology and the digital transformation of financial services, together with monetary policy and payments. Fintech and digital finance form a basic theme, while digital payments, India and the cashless society are niche topics.



1 (motor): digital transformation in financial services, monetary policy, monetary policy and economic impact, banking stability, regulation, efficiency

2 (motor): blockchain technology applications and security, digital transformation in law, digital ruble, cryptocurrency

3 (basic): fintech, crowdfunding, digital finance, blockchain, fintech, financial regulation

4 (emerging or declining): global financial crisis and policies, international payments, stablecoin, monetary sovereignty

5 (niche): financial inclusion, digital payments, india, cashless society

Figure 3. Thematic map of the keywords of research by economists

From our analysis, we observe a shift in the way economists discuss CBDC from 2014 to 2026. The early literature treated CBDC as a monetary policy instrument: half of the papers published before 2020 deal with monetary policy and its economic impact, against a fifth today. The topic model shows the same trend, as the topic on interest rates and uncertainty decreases by 9.6 points. Early papers also often discussed CBDC together with private cryptocurrencies: the largest topic of the period is characterized by words such as *cryptocurrencies* and *private*, and it is also the topic that lost the most ground since then (−13.2 points).

After 2020, as central banks started to work on their own projects, the discussion became more national. The digital euro and the digital ruble, almost absent before 2020, now appear in 11% and 15% of the papers, and *euro*, *ruble* and *India* are among the words that best characterize recent years. At the same time, financial inclusion is the theme that grows the fastest, from 6% to 16% of the papers. Overall, the literature moved from general questions on the effects of a CBDC to questions on specific national projects and on who they are meant for.

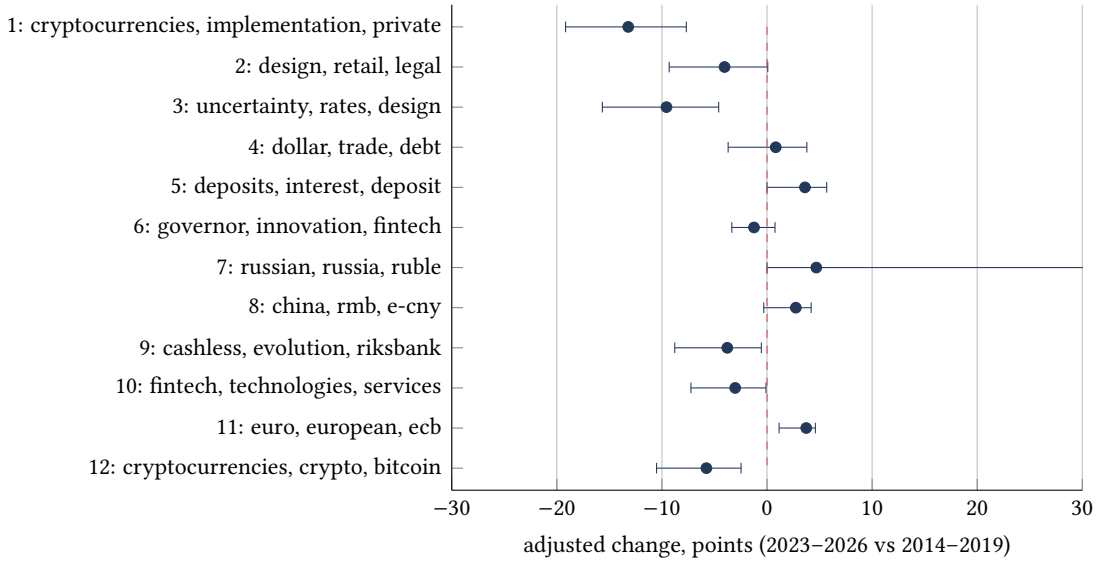


Figure 4. Change in the share of the twelve largest topics, adjusted for the type of author and the language, with 95% confidence intervals (axis cut at ±30 points: topics almost absent in 2014–2019 have very wide intervals)

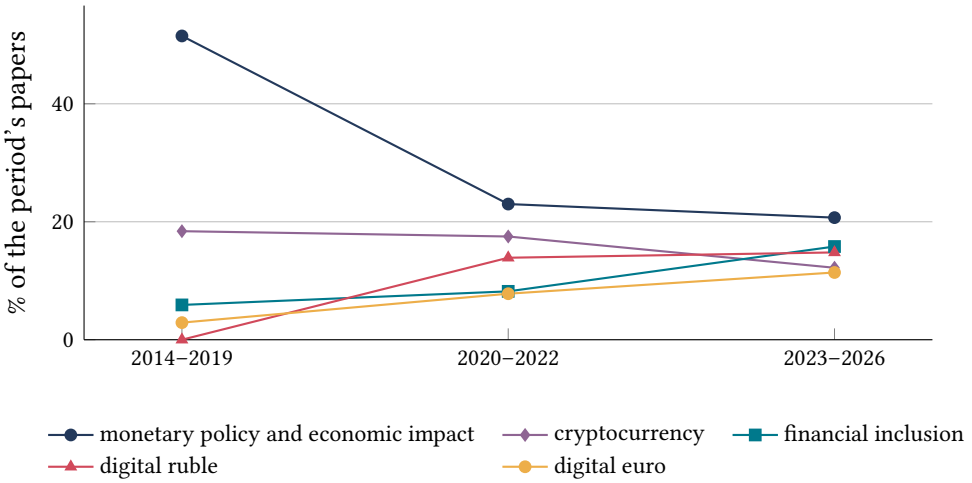


Figure 5. Share of research papers carrying each keyword, by period (136 papers in 2014–2019, 755 papers in 2020–2022, 2,919 papers in 2023–2026)

6 How economists discuss CBDCs: tone and stance

6.1 Method: tone (FinBERT) and stance (reading and a local language model)

We analyze both tone and stance, as they capture two different aspects of the discussion around CBDCs. Tone measures whether the language used to discuss CBDCs is positive or negative, whereas stance captures the author's overall position toward CBDCs for instance whether the text is favorable, neutral, or critical. For the tone analysis, we keep the English sentences that explicitly mention a CBDC and apply *FinBERT*, a model trained on financial text. For each sentence, the model gives three probabilities: $P(+)$, $P(0)$ and $P(-)$. We define the tone of sentence s and of document d as

$$\text{Tone}_s = P_s(+) - P_s(-) \in [-1, 1], \qquad \text{Tone}_d = \frac{1}{S_d} \sum_{s \in d} \text{Tone}_s, \tag{12}$$

where $P_s(+)$ and $P_s(-)$ are the probabilities that sentence s is positive and negative, and S_d is the number of CBDC sentences in document d. A tone of 1 means certainly positive, -1 certainly negative and 0 neutral or balanced. The scores are first averaged within each document and then across groups and years. This avoids giving too much weight to longer texts. In order to check the quality of the measure, 90 sentences from economists' texts were also coded by hand. The agreement between FinBERT and manual coding gives a Cohen's κ of 0.52.

For stance, we use two approaches. First, texts are read manually to estimate the level of favourable, neutral and critical positions. The share of favourable texts is reported with a Wilson confidence interval:

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \tag{13}$$

where n is the number of texts read in the group, $\hat{p} = k/n$ is the share of them, k, judged favourable, and $z = 1.96$ is the standard normal quantile for a 95% interval. Unlike the usual interval $\hat{p} \pm z\sqrt{\hat{p}(1-\hat{p})/n}$, it stays within $[0, 1]$ and remains reliable for small samples and shares close to 0 or 1.

Manual reading gives a more reliable measure of the level of support, however it cannot easily be applied to the whole corpus. To study changes over time, we therefore use a local *Qwen2.5 7B* model on random samples of texts for each actor and period. The change in stance is measured as:

$$\Delta\hat{p} = \hat{p}_{\text{late}} - \hat{p}_{\text{early}}, \tag{14}$$

where $\hat{p}_{\text{early}}$ and $\hat{p}_{\text{late}}$ are the shares of texts labelled favourable by the model among the n_{early} texts of 2014–2019 and the n_{late} texts of 2023–2026 of the sample (up to 100 per actor and period), with an approximate 95% confidence interval given by:

$$\Delta\hat{p} \pm 1.96\sqrt{\frac{\hat{p}_{\text{late}}\,(1-\hat{p}_{\text{late}})}{n_{\text{late}}} + \frac{\hat{p}_{\text{early}}\,(1-\hat{p}_{\text{early}})}{n_{\text{early}}}}. \tag{15}$$

The local LLM reaches a Cohen's κ of 0.45 against manual coding, but it tends to classify too many texts as favourable. For that reason, we use manual reading for the level of stance, and the LLM mainly to compare changes across periods

6.2 Findings

The analysis reveals a clear increase in the positivity of CBDC-related discourse over time. Among research economists, average net tone rises from approximately 0.07 in 2019 to 0.25 in 2025, while central-bank and institutional texts remain slightly more positive throughout the period. Overall, the average net tone is 0.19 for research economists and 0.22 for central banks and institutions. This suggests that the language used to discuss CBDCs has become progressively more positive in both groups, although institutional discourse is consistently somewhat more favourable in tone.

However, the stance analysis shows a stronger difference between the two groups. In the manually coded sample, research economists are predominantly neutral: only 8% of texts are classified as favourable and 2% as critical. By contrast, 52% of central-bank and institutional texts are favourable, while 6% are critical. This indicates a difference in the role played by each group in the CBDC debate: academic research is mainly analytical, whereas central banks and institutions are more likely to express explicit support for CBDCs.

The evolution over time points in the same direction. The local language model reveals that the share of favourable research texts rises from 26% in 2014–2019 to 40% in 2023–2026, corresponding to an increase of 14 percentage points. The increase is substantially larger for central banks and institutions, from 25% to 65%, or 40 percentage points. Since the LLM tends to overestimate the absolute level of favourable stance, these percentages should not be interpreted as precise levels of support, rather, they indicate the direction and magnitude of change across periods.

Overall, the results point to a gradual change in the way CBDCs are discussed. The language becomes more positive in both groups, but the shift towards an explicitly favourable stance is much stronger among central banks and institutions. Research economists remain more cautious and analytical, while institutional actors increasingly express support for CBDCs as the debate develops.

6.3 Do policy events move the tone?

To assess whether major policy developments are associated with sudden changes in CBDC discourse, we compare the average tone of texts published in the 180 days before and after each of twelve major CBDC-related events. The events include, for instance, the Facebook's announcement of Libra, the ECB's digital euro report, the launch of the Sand Dollar and eNaira, and the EU digital euro proposal. For each event, the change in tone is defined as the difference between the post-event and pre-event averages.

To determine whether the observed changes are unusually large, we compare them with changes calculated around 500 randomly selected dates. This gives us a simple comparison point: if the change around a real event is not larger than what we often observe around random dates, it is difficult to argue that the event itself produced a clear shift in tone.

The results provide no evidence of a discrete change in tone around any of the twelve events. None of the estimated changes is statistically significant, and the smallest p-value equal to 0.24. For instance, the announcement of Libra is associated with a decline in tone of -0.04, while the launch of the eNaira is followed by an increase of $+0.05$, yet neither change is unusual relative to the placebo distribution.

These findings suggest that the increasingly positive tone documented in the previous section is gradual rather than driven by one specific policy announcement or launch. Instead, the change seems to be gradual. Thus, the evolution of CBDC discourse appears to move alongside the broader policy environment. Policy developments may still matter, but their effect is more likely to build up over time rather than appear as an immediate jump in tone.

7 Why: stated motivations

7.1 Method: a validated codebook

To identify the main motivations used for study and develop CBDCs, we construct a rule-based codebook covering ten motivations. The objective is to capture motivations that are explicitly stated by the authors, rather than simply detecting whether a topic appears somewhere in the text.

The codebook was built from a development set of 100 introductions. For each motivation, we identified the expressions authors use to state it and translated these expressions into regular-expression in seven languages. The resulting rules include, for example, expressions related to *financial inclusion*, such as "*unbanked*" or "*access to financial services*", and to *monetary sovereignty*, such as "*global stablecoin*", "*Libra*", or "*dollarization*". The final codebook was then applied to 4,110 texts.

The codebook was then evaluated on a separate test set of 74 texts, which was not used during its construction. These texts were coded manually, and performance was assessed on the 50 documents written by economists. We use three main validation measures, computed on the $50 \times 10 = 500$ (text, motivation) pairs: for each pair, the codebook and the reader each say whether the text states the motivation. A true positive (TP) is a pair where both say yes, a false positive (FP) a pair where only the codebook says yes, and a false negative (FN) a pair where only the reader says yes. Precision captures the share of detected motivations that are correctly identified:

$$\text{Precision} = \frac{TP}{TP + FP}, \tag{16}$$

while recall measures how many of the motivations identified by human reading are recovered by the codebook:

$$\text{Recall} = \frac{TP}{TP + FN}. \tag{17}$$

Finally, Cohen's κ measures agreement between the codebook and manual coding beyond what would be expected by chance:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \qquad p_e = q_r\, q_c + (1 - q_r)(1 - q_c), \tag{18}$$

where p_o is the observed agreement, the share of pairs on which the codebook and the reader give the same answer, and p_e the agreement expected by chance, with q_r and q_c the shares of pairs coded yes by the reader and by the codebook. $\kappa = 1$ means perfect agreement and $\kappa = 0$ no more agreement than chance.

On the held-out economists' sample, the codebook reaches a precision of 0.70, a recall of 0.41, and a Cohen's κ of 0.46. We also compare its performance with several machine-based alternatives, including *SBERT classifiers*, an *NLI model* and local *Qwen2.5* language models, using the same validation texts.

7.2 Findings

The results show that the reasons given for discussing CBDCs have changed over time. In the first period, monetary policy was one of the main motivations, appearing in 36.8% of texts in 2014–2019. This share falls to 24.4% in 2023–2026, which corresponds to an adjusted decrease of 15.1 percentage points. The decline of cash also becomes less important over time, with an estimated decrease of 6.5 points.

By contrast, several motivations become more prominent. Financial inclusion rises from 3.8% of texts in 2014–2019 to 16.8% in 2023–2026, corresponding to an adjusted increase of 12.3 points. Geopolitics and sanctions show a similarly strong increase of 12.4 points, while cross-border payments increase by

8.7 points. In addition to that, payment efficiency and privacy and surveillance become more frequent motivations.

Overall, these results suggest that the way CBDCs are justified has gradually changed. Earlier discussions were more focused on monetary policy and the decline of cash, while recent texts place more emphasis on financial inclusion, payments and geopolitical issues.

There are also some differences between research economists and central banks. Central-bank and institutional texts are less likely than research papers to frame CBDCs through monetary policy, technology, financial inclusion or cross-border payments, while the difference for geopolitics and sanctions is close to zero.

Given the relatively low recall of the codebook, these percentages should not be read as exact measures of prevalence. The main interest lies in the direction and relative magnitude of the changes across periods rather than in the absolute levels themselves.

8 When: the policy environment

8.1 Method: bursts, event study with placebo dates, difference-in-differences

To examine how the CBDC discussion evolves alongside its policy environment, we use three complementary methods. First, we use *Kleinberg's burst detection algorithm* to identify periods in which some keywords become much more frequent. For keyword w in quarter t, r_t is the number of papers using the keyword among the d_t research papers of that quarter. The model compares a normal state, with rate $p_0 = R/D$, where $R = \sum_t r_t$ and $D = \sum_t d_t$ are the totals over all quarters, to a burst state, with a higher rate $p_1 = 2p_0$. The cost of being in state $i \in \{0, 1\}$ in quarter t is minus the log-likelihood of the observed count,

$$\sigma_i(t) = -\ln\left[\binom{d_t}{r_t} p_i^{r_t} (1 - p_i)^{d_t - r_t}\right], \tag{19}$$

and moving from the normal to the burst state costs an extra $\ln Q$, where Q is the number of quarters, while leaving a burst is free. Since entering a burst has a cost, short random fluctuations are less likely to be selected. The final sequence of normal and burst periods, the one with the lowest total cost, is chosen using the *Viterbi algorithm*.

Second, we run an event study with placebo dates to see whether attention to CBDCs increases immediately after major policy events. We estimate a baseline publication rate over the period from 90 to 15 days before each event, $\lambda = N_{\text{pre}}/76$, where N_{pre} is the number of texts dated to the day published in these 76 days. The expected number of texts in the 15 days starting on the day of the event is then:

$$E = 15\lambda \tag{20}$$

and the observed increase is measured using:

$$z = \frac{N - E}{\sqrt{E}}, \tag{21}$$

where N is the number of texts actually published in these 15 days. If publications followed a Poisson process with rate λ, z would be approximately standard normal. Because publication activity is not stable over time, each event is compared with 500 random placebo dates. The p-value is the share of placebo dates with a larger z-score than the actual event.

Finally, we use a difference-in-differences approach to examine whether geopolitical motivations became more important after 2021, especially in texts related to Russia. The outcome variable indicates

whether a motivation is stated, while the treatment group includes texts about Russia or written in Russian:

$$y_{it} = \alpha + \eta\, T_i + \mu_t + \sum_{k \neq 2021} \gamma_k T_i \mathbf{1}[t = k] + \varepsilon_{it}, \tag{22}$$

where y_{it} equals 1 if text i, published in year t (2018–2026), states the motivation and 0 otherwise; T_i equals 1 if text i is in the treatment group and 0 otherwise; α is the intercept, η the permanent gap between the two groups, μ_t a year effect common to all texts ($\mu_{2021} = 0$), $\mathbf{1}[t = k]$ equals 1 if $t = k$, and ε_{it} is the error term. The model is a linear probability model estimated by ordinary least squares with robust (HC1) standard errors. The coefficients γ_k measure how the gap between Russia-related texts and the rest of the corpus changes in each year relative to 2021. This allows us to see whether the post-2021 shift is stronger in Russia-related discourse than in the broader CBDC literature. The coefficients for the years before 2021 test whether the two groups followed parallel trends before the shock. The single effect reported in the findings comes from the same model in which the yearly interactions are replaced by one term, $\gamma\, T_i \mathbf{1}[t > 2021]$.

8.2 Findings

The results show that the focus of the CBDC debate changes quite clearly over time. In the earlier years, the main bursts are related to monetary policy, economic models and economic impact. More recent bursts are instead linked to India, monetary sovereignty and the digital euro. This suggests a shift from a more theoretical discussion toward topics that are more closely connected to specific countries and policy developments.

Besides, policy events also seem to matter for the amount of attention given to CBDCs, although not in every case. Out of the twelve events considered, four are followed by a rise in publication activity that is larger than 95% of the placebo dates. The strongest increases are observed around the ECB preparation phase and the launch of the Sand Dollar. At the same time, several other major events do not generate a noticeable jump in attention.

The results on Russia give a similar picture. After 2021, texts related to Russia became more likely to mention geopolitics and sanctions as a motivation for CBDCs. The increase is about 11.6 percentage points larger than in the rest of the corpus, and the gap becomes wider over time.

Overall, the results suggest that the CBDC discussion evolves together with changes in the policy and geopolitical environment. Some events create short-term peaks in attention, while broader geopolitical developments seem to be associated with changes in the way CBDCs are discussed. However, these results should not be read as direct proof of causality

9 The discussion and its environment

In this section, we put together the previous results period by period, and we discuss what they allow us to say, or not, about the causes of the changes we observe.

9.1 Period by period

From our analysis, we observe that between 2014 and 2019, economists wrote little about CBDCs, with 53 texts a year on average. The discussion was mainly about monetary policy: it is the most frequent motivation, and keywords such as monetary policy and economic theories suddenly become much more frequent during this period. The main event of this period, the announcement of Facebook's project Libra in June 2019, is not followed by any rise in attention. In addition, the tone of the texts remains cautious: the net tone is only 0.06 for research economists and 0.10 for central banks.

Between 2020 and 2022, the number of texts increases to 324 a year. This is the period corresponding to the first central bank reports and launches, but only two of the seven events of the period are followed by a significant rise in attention: the ECB report on a digital euro and the launch of the Sand Dollar in the Bahamas, both in October 2020. At the same time, the discussion becomes more favorable, especially among central banks, whose net tone rises to 0.22.

Finally, since 2023, economists write about 840 texts a year, and the discussion is now about national projects. Out of the four events of this period, only two lead to more texts: the start of the ECB preparation phase and, just barely, the EU bill on the digital euro. Keywords such as India, its Unified Payments Interface and monetary sovereignty also become much more frequent. This is also the period when texts are the most positive about CBDCs: compared with 2014-2019, the share of texts in favor of CBDCs rose by 40 points for central banks and by 14 points for research economists. In short, economists moved from a theoretical question on monetary policy to real national projects and became more positive as these projects moved forward.

9.2 What can and cannot be said about causes

Our analysis is purely descriptive: it allows us to see what changes and what moves at the same time, but not to prove what causes these changes. To see how far we can go, we test three links. The first one is between events and attention: 4 of the 12 policy events are followed by more texts than 95% of random dates. In other words, economists write more after some events, but this only tells us about timing and not about causes. Then, by testing the link between events and tone, we observe that none of the 12 events changes the tone of the texts: the tone rises slowly over the whole period, without any sudden change after a specific event. Finally, we observe that after the sanctions on Russia in 2022, the share of texts about Russia, or written in Russian, that mention geopolitics and sanctions rose by 11.6 points more than in other texts. However, three other motivations also changed in these texts (financial inclusion, financial stability and monetary policy), so this result fits with an effect of the sanctions but does not prove it.

10 Robustness

In order to test whether our analysis really shows the economists' debate and is not driven by the way we built the data, we test the robustness of our main results.

To build our corpus and our measures, we made several choices that could affect the results. For example, the fall of monetary policy could simply come from the growing share of texts written in Russian, which may discuss CBDCs differently, and not from a real change in the debate. Therefore, to check for robustness, we run the same calculations on six different samples in which we change one choice at a time:

- only texts where CBDC is the main theme.
- only texts in English, since the mix of languages changes over time.
- only journal articles for research, since working papers may differ from published work.
- only texts before 2026.
- a less strict coding of motivations, since our codebook misses many of them.
- research from every discipline, to check that our filter on economists does not create the results by itself.

Table 3. Main results under alternative samples

		Motivations: change, points		Stance	Events
Sample	**Texts**	**Inclusion**	**Monetary policy**	**CB – research**	**significant**
Baseline: economists	4,710	+12.3 [+8, +15]	−15.1 [−22, −8]	+16	4/12
Main theme only	3,513	+14.9 [+10, +18]	−16.0 [−25, −7]	+13	4/12
English only	3,721	+13.7 [+9, +16]	−16.0 [−23, −7]	+16	4/12
Journal articles only	3,171	+12.3 [+6, +15]	−16.8 [−25, −8]	+11	4/12
Without 2026 (partial year)	3,547	+11.8 [+7, +15]	−15.3 [−23, −8]	+15	4/12
High-recall motivation coding	4,710	+15.6 [+10, +19]	−8.8 [−16, −2]	+16	4/12
All disciplines	8,213	+13.2 [+10, +15]	−15.0 [−21, −9]	+15	4/12

Motivations: change between 2014–2019 and 2023–2026, adjusted for period, type of author and language (logit, 95% confidence interval). Stance: gap in the share of favourable texts between central banks and research economists (LLM labels, points). Events: number of the 12 policy events followed by more texts than 95% of random dates.

The results we obtained seem to be very stable. Across all samples, the rise of financial inclusion lies between +11.8 and +15.6 points and the fall of monetary policy between −8.8 and −16.8 points, and both changes are statistically significant in every sample. With English texts only, the fall of monetary policy is almost unchanged (−16.0 points), so it does not come from the growing share of Russian texts. It is smaller with the less strict coding of motivations, but still significant. Central banks are also more favourable to CBDCs than research economists in every sample, by 11 to 16 points, the gap in tone is always small (0.01 to 0.03), and the same four events are followed by a rise in attention each time. In short, our main results do not depend on the way we built the corpus. However, these tests only tell us that they are stable, not what causes them.

11 Discussion

From our analysis, we observe that economists increasingly see CBDCs less as a monetary policy instrument and more as a tool for inclusion, payments and geopolitics. Monetary policy, central in the early literature lost ground, while financial stability remained a constant concern. Monetary sovereignty also stays stable, while geopolitics and sanctions are one of the motivations that grow the most. Indeed, the international dimension of CBDCs, through geopolitics and cross-border payments, has become much more important.

Central banks also play a key role in this debate. Economists from central banks and international institutions write 10% of the research papers but receive 48% of the citations, and central bank texts are much more favourable to CBDCs than those of research economists, who remain mostly neutral. This means that the most cited work on CBDCs comes largely from institutions that are directly involved in the policy debate, and not only from independent research.

11.1 Limitations

However, our study also highlights some limitations. First, we identify economists by the field of their texts and the institution of their authors, and not by their education. Some economists who publish in other fields may therefore be missed, while some non-economists may be kept. Then, our measures of tone only cover part of the debate since it is measured on English texts only, which represent 79% of the corpus. Therefore, it mainly reflects the English-speaking debate, while the share of texts in other languages, like Russian ones, grows over time. Finally, our analysis is descriptive: it shows how the debate changes, but not why. For example, explaining why central banks became more favourable to CBDCs would require other data, such as interviews with economists or a study of their careers.

12 Conclusion

References

Agur, I., A. Ari, and G. Dell'Ariccia (2019). *Designing Central Bank Digital Currencies*. IMF Working Paper 19/252. International Monetary Fund.

Andolfatto, D. (2020). Assessing the Impact of Central Bank Digital Currency on Private Banks. *The Economic Journal* 131.634. Published online 2020, issue dated 2021, pp. 525–540. DOI: 10.1093/ej/ueaa073.

Araci, D. (2019). *FinBERT: Financial Sentiment Analysis with Pre-Trained Language Models*. arXiv: 1908.10063.

Aria, M. and C. Cuccurullo (2017). bibliometrix: An R-Tool for Comprehensive Science Mapping Analysis. *Journal of Informetrics* 11.4, pp. 959–975. DOI: 10.1016/j.joi.2017.08.007.

Auer, R., J. Frost, L. Gambacorta, C. Monnet, T. Rice, and H. S. Shin (2022). Central Bank Digital Currencies: Motives, Economic Implications, and the Research Frontier. *Annual Review of Economics* 14, pp. 697–721. DOI: 10.1146/annurev-economics-051420-020324.

Barrdear, J. and M. Kumhof (2016). *The Macroeconomics of Central Bank Issued Digital Currencies*. Staff Working Paper 605. Bank of England.

Bindseil, U. (2020). *Tiered CBDC and the Financial System*. ECB Working Paper 2351. European Central Bank.

Blondel, V. D., J.-L. Guillaume, R. Lambiotte, and E. Lefebvre (2008). Fast Unfolding of Communities in Large Networks. *Journal of Statistical Mechanics: Theory and Experiment*, P10008. DOI: 10.1088/1742-5468/2008/10/P10008.

Bordo, M. D. and A. T. Levin (2017). *Central Bank Digital Currency and the Future of Monetary Policy*. NBER Working Paper 23711. National Bureau of Economic Research. DOI: 10.3386/w23711.

Bradford, S. C. (1934). Sources of Information on Specific Subjects. *Engineering* 137, pp. 85–86.

Brunnermeier, M. K. and D. Niepelt (2019). On the Equivalence of Private and Public Money. *Journal of Monetary Economics* 106, pp. 27–41. DOI: 10.1016/j.jmoneco.2019.07.004.

Callon, M., J.-P. Courtial, and F. Laville (1991). Co-Word Analysis as a Tool for Describing the Network of Interactions between Basic and Technological Research: The Case of Polymer Chemistry. *Scientometrics* 22.1, pp. 155–205. DOI: 10.1007/BF02019280.

Campello, R. J. G. B., D. Moulavi, and J. Sander (2013). "Density-Based Clustering Based on Hierarchical Density Estimates". In: *Advances in Knowledge Discovery and Data Mining (PAKDD 2013)*. Springer, pp. 160–172. DOI: 10.1007/978-3-642-37456-2_14.

Chaum, D. (1983). "Blind Signatures for Untraceable Payments". In: *Advances in Cryptology: Proceedings of Crypto 82*. Springer, pp. 199–203. DOI: 10.1007/978-1-4757-0602-4_18.

Cobo, M. J., A. G. López-Herrera, E. Herrera-Viedma, and F. Herrera (2011). An Approach for Detecting, Quantifying, and Visualizing the Evolution of a Research Field: A Practical Application to the Fuzzy Sets Theory Field. *Journal of Informetrics* 5.1, pp. 146–166. DOI: 10.1016/j.joi.2010.10.002.

Cohen, J. (1960). A Coefficient of Agreement for Nominal Scales. *Educational and Psychological Measurement* 20.1, pp. 37–46. DOI: 10.1177/001316446002000104.

Diamond, D. W. and P. H. Dybvig (1983). Bank Runs, Deposit Insurance, and Liquidity. *Journal of Political Economy* 91.3, pp. 401–419. DOI: 10.1086/261155.

Eck, N. J. van and L. Waltman (2009). How to Normalize Cooccurrence Data? An Analysis of Some Well-Known Similarity Measures. *Journal of the American Society for Information Science and Technology* 60.8, pp. 1635–1651. DOI: 10.1002/asi.21075.

Grootendorst, M. (2022). *BERTopic: Neural Topic Modeling with a Class-Based TF-IDF Procedure*. arXiv: 2203.05794.

Kleinberg, J. (2003). Bursty and Hierarchical Structure in Streams. *Data Mining and Knowledge Discovery* 7.4, pp. 373–397. DOI: 10.1023/A:1024940629314.

Lotka, A. J. (1926). The Frequency Distribution of Scientific Productivity. *Journal of the Washington Academy of Sciences* 16.12, pp. 317–323.

McInnes, L., J. Healy, and J. Melville (2018). *UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction*. arXiv: 1802.03426.

Monroe, B. L., M. P. Colaresi, and K. M. Quinn (2008). Fightin' Words: Lexical Feature Selection and Evaluation for Identifying the Content of Political Conflict. *Political Analysis* 16.4, pp. 372–403. DOI: 10.1093/pan/mpn018.

Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System*. URL: https://bitcoin.org/bitcoin.pdf.

Qwen Team (2024). *Qwen2.5 Technical Report*. arXiv: 2412.15115.

Reimers, N. and I. Gurevych (2019). "Sentence-BERT: Sentence Embeddings Using Siamese BERT-Networks". In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pp. 3982–3992. DOI: 10.18653/v1/D19-1410.

Roberts, M. E., B. M. Stewart, D. Tingley, C. Lucas, J. Leder-Luis, S. K. Gadarian, B. Albertson, and D. G. Rand (2014). Structural Topic Models for Open-Ended Survey Responses. *American Journal of Political Science* 58.4, pp. 1064–1082. DOI: 10.1111/ajps.12103.

Tobin, J. (1987). The Case for Preserving Regulatory Distinctions. *Challenge* 30.5, pp. 10–17. DOI: 10.1080/05775132.1987.11471321.

A Search vocabulary

A single vocabulary drives both the searches and the relevance score (Section 3). It has 31 concepts, each written as a case-insensitive regular expression, and 115 search phrases derived from them, in 12 languages. Three principles guided it.

1. **Recall first.** The vocabulary covers the generic term in every major language, the names of national projects, the BIS Innovation Hub experiments, and the older names used before "CBDC" became standard (*digital fiat currency*, *central bank cryptocurrency*, *Fedcoin*, *RSCoin*, *CADcoin*, *digital base money*).

2. **Strong and ambiguous terms.** A *strong* term identifies a central bank digital currency on its own (*central bank digital currency*, *digital euro*, *e-CNY*). An *ambiguous* term does not: "CBDC" is also *chronic bullous dermatosis of childhood*, "DCEP" a chemical compound, "Project Stella" a satellite. Ambiguous terms count only when the document also contains monetary context: *currency, monetary, payment, central bank, banking, ledger, stablecoin*, and their equivalents in French, Spanish, Portuguese, German, Italian, Russian, Chinese, Japanese and Korean. When the only evidence is in the body of the text, a stricter context list is required (*central bank, digital currency, monetary, payments*).

3. **Never-scoring terms.** *Digital dollar* and *Sand Dollar* are reported but never scored: they are too often used for private tokens or unrelated products.

Table 4. Concepts of the search vocabulary (selection)

Concept	Forms matched
Generic	CBDC, wCBDC, rCBDC, mCBDC; central bank(s) digital currency/currencies; digital central bank money; tokenised central bank money or reserves
Other languages	monnaie numérique de banque centrale (MNBC); moneda / moeda digital de banco central; digitales Zentralbankgeld; valuta / moneta digitale di banca centrale; Japanese, Korean and Russian equivalents (*tsifrovaya valyuta tsentral'nogo banka*)
Euro area	digital euro, euro numérique, digitaler Euro, euro digitale, euro digital
China	e-CNY, digital yuan / renminbi, DC/EP (Digital Currency Electronic Payment), Chinese forms
Other projects	digital pound (Britcoin), e-krona, digital / e-rupee, eNaira, DCash, JAM-DEX, digital ruble (*tsifrovoy rubl'*), Drex / real digital, e-hryvnia, digital yen, e-HKD, e-peso, Bakong, e-Cedi, digital shekel, lira, won, rial, e-dirham
Early names	digital fiat currency, central bank cryptocurrency, central bank e-money / electronic money / digital cash, sovereign or national digital currency, digital legal tender, Fedcoin, RSCoin, CADcoin
BIS projects	mBridge; Projects Helvetia, Jura, Dunbar, Icebreaker, Aurum, Rosalind, Tourbillon, Mariana, Hamilton, Stella, Ubin, Inthanon, Khokha, Jasper, Agorá, Orchid, Polaris, Sela, Mandala, Ensemble
Adjacent	tokenised deposits, unified ledger (ambiguous: counted only with monetary context)

The full list of regular expressions and search phrases is in `cbdc_biblio/config.py`.

B Harvesting algorithms

B.1 Sources

Records come from four families of sources (Table 5). Nothing is filtered at harvest time: every record found by a search is kept, and relevance is decided afterwards on the merged corpus. Every step writes a checkpoint, so an interrupted harvest resumes where it stopped.

Academic databases. OpenAlex (journal articles, working papers, books, chapters, preprints, reports), Semantic Scholar, arXiv, HAL and DOAJ. Each search phrase is queried in titles and abstracts; results are merged on the source's identifier. OpenAlex reference lists are resolved into citation strings, which the co-citation analysis uses.

Central bankers' speeches. The BIS bulk archive (all central banks, 1996 onwards, full text), the ECB's speech dataset (full text, including non-English originals), and the BIS RSS feed for speeches published after the bulk archive.

Authorities. A crawl starting from 26 CBDC pages of central banks, EU institutions and international organisations, following links one level deep within the same site, PDFs included.

Press. Google News, searched through its RSS interface in 55 combinations of country edition, language and query. GDELT was also queried but throttled the connection for hours after a few requests, so it contributed nothing to the final corpus.

Table 5. Records by source and filtering stage

Source	Collected	CBDC a theme	By economists
Google News	24,309	9,496	0
OpenAlex	20,596	7,266	4,043
Semantic Scholar	12,528	4,000	2,164
DOAJ	1,750	485	251
BIS central bankers' speeches	618	465	465
Central banks and authorities (crawl)	266	239	205
HAL	152	83	31
ECB speeches	127	108	108
arXiv	93	90	24
Distinct records	48,188	17,709	4,710

A record found in several sources counts once per source; the last row counts distinct records.

B.2 Pagination and rate limits

Academic APIs return results page by page; the harvester follows the API's cursor (200 records per page for OpenAlex) until no page is left. Requests are spaced by a minimum interval Δ,

$$t_{k+1} - t_k \geq \Delta,$$

with $\Delta = 0.12$ s for OpenAlex, 3.5 s for arXiv and 1.2 s for Google News. After the j-th consecutive refusal (HTTP 429 or 5xx), the harvester waits $w_j = \min(60j, 900)$ seconds before retrying.

B.3 Press: recursive bisection of time windows

A Google News search returns at most about 100 items, whereas a single month can hold thousands of articles. Each search is therefore run on time windows that are split until every window falls below the cap:

1. Start from monthly windows $W = [a, b)$ covering January 2000 to the harvest date, on a stack.
2. Pop a window and fetch the items published in it.
3. If it returns $N(W) \geq 95$ items and lasts more than one day, split it at $m = a + (b - a)/2$, push $[m, b)$ and $[a, m)$, and go back to step 2.

4. Otherwise save the items and checkpoint the window.

A month is split at most $\lceil \log_2 31 \rceil = 5$ times, and keeping L windows costs $2L - 1$ requests. Google News limits requests over time, and the harvest was stopped with about 15 of the 55 searches unfinished; press counts are therefore incomplete, and skewed towards recent years.

B.4 Full text, enrichment and de-duplication

Full text. PDFs are extracted with PyMuPDF and web pages with trafilatura, which also recovers the page's date, author and outlet when the record lacks them. Texts are split into introduction, conclusion and section headings; for speeches and press, the opening and closing paragraphs play these roles.

Enrichment. Language is detected with `langdetect`; keywords are extracted with YAKE (one to three words) when a record has no author or indexed keywords. Dates are kept with their precision (day, month, year), and 1 January dates are flagged, since they often stand for a year only.

De-duplication. Records are matched on a key: the DOI for academic records, else their normalised title and year; the date and normalised title for speeches; the URL for other records. A second pass merges academic records with the same normalised title and year but different DOIs (a preprint and its published version, for instance). Merged records keep the most precise date, the longest abstract and the union of keywords and sources.

B.5 Relevance score

Each available field f of a record receives a centrality $c_f \in [0, 1]$: 1 for a title containing a strong term; 0.4 for a keyword list containing one (keywords are partly machine-assigned); for the abstract, the share of its sentences that contain one; for the body, half the density of mentions (per 1,000 words, saturating at 3) and half their spread (the share of ten equal slices of the text that contain one). The score is

$$\text{score} = 100\Big(0.6 \max_{f \in F} c_f + 0.4 \frac{1}{|F|} \sum_{f \in F} c_f\Big),$$

where F is the set of fields available, so a missing abstract or full text is not penalised. Records scoring at least 50 are labelled *main theme*, from 25 to 49 *secondary theme*, and below 25 *mentioned only*. Records whose only matches are ambiguous terms without monetary context are *ambiguous terms only*; records without any match are *off-topic*; press items whose text could not be retrieved are *unverified*. Records with CBDC as a theme but dated before 2014 were all homonyms or misdated, and are labelled *dated before 2014*. Table 6 gives the labels and their validation.

Table 6. Relevance labels and their validation

Label	Records	Read	Theme	95% CI
main theme	15,464	30	29	0.83–0.99
secondary theme	2,245	29	11	0.23–0.56
mentioned only	489	30	1	0.01–0.17
unverified (no text)	13,446	19	2	0.03–0.31
ambiguous terms only	1,709	10	3	0.11–0.60
off-topic	14,803	20	0	0.00–0.16
dated before 2014	32	0	0	–

Read: records of the stratified sample (138 texts) read blind to the label; Theme: read as having CBDC as a theme; Wilson intervals on the share.

B.6 Who counts as an economist

For each academic record with CBDC as a theme, the OpenAlex field of its primary topic and the institutions of its authors were retrieved (`scripts/tag_economists.py`). A text is attributed to economists when it is research classified in *Economics, Econometrics and Finance*, or research with an author at a central bank, the IMF or the BIS, or, for records absent from OpenAlex, research published in an economics venue; when it is a central bank speech; or when it is a document of a central bank or an international financial institution. Tables 7 and 8 show the result.

Table 7. Who counts as an economist

Texts with CBDC as a theme	Texts	
Research classified in Economics, Econometrics and Finance	3,783	kept
Research in another field, with a central bank, IMF or BIS author	148	kept
Research without OpenAlex record, in an economics venue	63	kept
Research in other fields	3,469	dropped
Central bank speeches	511	kept
Documents of central banks and international institutions	205	kept
Documents of governments and parliaments	34	dropped
Press articles	9,496	dropped
Texts by economists	4,710	

Field: OpenAlex primary topic of the work; author institutions from OpenAlex and from the record's affiliations.

Table 8. Research with CBDC as a theme left out by the economist filter, by field

OpenAlex field of research dropped	Texts
Computer Science	1,769
Business, Management and Accounting	663
no OpenAlex record	473
Social Sciences	281
Decision Sciences	244
Engineering	14
Medicine	7
Health Professions	5
Environmental Science	4
Arts and Humanities	3
Agricultural and Biological Sciences	2
Neuroscience	2
Immunology and Microbiology	1
Materials Science	1

C Validation samples and coding guide

Every measure used in the paper was compared with a reading of the texts. The samples were drawn at random; the motivation, stance and tone samples were topped up with texts written by economists so that each holds at least 50 of them (Table 9). Coding was done blind to the machine labels, by one coder assisted by a language model.

C.1 Coding guide

Relevance. Is a central bank digital currency the theme of the text, or one of its main themes? A passing mention, a homonym, or a text about cryptocurrencies or stablecoins only is coded *no*.

Table 9. Hand-coded validation samples

Sample	Units	From economists' texts
Relevance	140	25
Motivations, development	100	60
Motivations, test (held out)	74	50
Stance	177	100
Tone (sentences)	136	90

Motivations. Unit: the first 1,500 characters of the introduction, else of the abstract, else of the text. A motivation is coded when the text raises it as a reason to study, issue or worry about a CBDC; a text can have several. The ten motivations are defined as follows.

- *Decline of cash*: cash use is falling, or access to cash or public money is at risk.
- *Financial inclusion*: reaching unbanked or underbanked people; access to financial services.
- *Monetary sovereignty*: a threat from private money, stablecoins, Big Tech, cryptocurrencies or foreign currencies to the central bank's control of money.
- *Payment efficiency*: cheaper, faster, more efficient, resilient or innovative domestic payments.
- *Cross-border payments*: international payments, remittances, cross-border settlement.
- *Monetary policy*: monetary policy, interest rates, transmission, money supply.
- *Financial stability*: bank disintermediation, bank runs, deposit outflows, risks to financial stability.
- *Privacy and surveillance*: privacy, anonymity, data protection, surveillance or control of users.
- *Geopolitics and sanctions*: geopolitics, sanctions, dollar dominance, internationalisation of a currency.
- *Technology*: blockchain, distributed ledgers, tokenisation, smart contracts, programmable money.

Stance. Unit: the title and abstract (research), or the title and the first five sentences that name a CBDC (speeches, documents). *Favourable*: supports CBDCs or presents them positively; *critical*: opposes them or stresses their risks and problems; *neutral*: descriptive or balanced. Analysing risks is neutral; arguing that they can be solved so that a CBDC can be issued, or recommending issuance, is favourable.

Tone. Unit: one English sentence that names a CBDC. *Positive* or *negative* when the sentence presents CBDCs or their effects favourably or unfavourably; *neutral* when it describes, defines or reports.

C.2 Results

The motivation codebook was written on the 100 development texts only and evaluated on the held-out test texts (Table 10). Each alternative coder was tested on the same texts (Tables 11 and 12); thresholds and classifiers were set on the development texts (motivations) or by 5-fold cross-validation (stance). Agreement is measured by Cohen's $\kappa = (p_o - p_e)/(1 - p_e)$, where p_o is the observed and p_e the chance agreement. The language model's stance labels are used only for changes over time: it labels too many texts favourable, a bias that cancels in a difference if it is stable. Stance levels come from reading (Table 14).

Table 10. Motivation codebook on the held-out texts written by economists

Motivation	Read	Coded	Precision	Recall	κ
Decline of cash	7	1	1.00	0.14	0.22
Financial inclusion	7	3	1.00	0.43	0.56
Monetary sovereignty	6	1	1.00	0.17	0.26
Payment efficiency	16	3	0.67	0.12	0.12
Cross-border payments	1	3	0.33	1.00	0.48
Monetary policy	7	9	0.56	0.71	0.55
Financial stability	10	7	0.57	0.40	0.37
Privacy and surveillance	3	2	1.00	0.67	0.79
Geopolitics and sanctions	7	4	1.00	0.57	0.70
Technology (DLT, programmability)	11	11	0.73	0.73	0.65
All (pooled)	75	44	0.70	0.41	0.46

50 held-out texts by economists (of 74), never used to write the codebook. Read: texts where reading found the motivation; Coded: texts flagged by the codebook.

Table 11. Candidate coders for motivations

Coder	P	R	F_1	κ	κ, all
Codebook (regex)	0.70	0.41	0.52	0.46	0.50
Codebook OR classifier	0.42	0.60	0.50	0.39	0.42
SBERT, classifier (100 labels)	0.36	0.37	0.37	0.25	0.28
Local LLM, Qwen2.5 7B	0.30	0.61	0.40	0.25	0.28
Local LLM, Qwen2.5 3B	0.31	0.23	0.26	0.15	0.16
SBERT, definitions (zero-shot)	0.21	0.71	0.32	0.12	0.15
NLI zero-shot (mDeBERTa)	0.22	0.39	0.28	0.11	0.14

Scores on the 50 texts written by economists; last column: κ on all 74 hand-coded texts. Thresholds and classifiers set on the 100 development texts.

Table 12. Candidate coders for stance

Coder	Accuracy	κ	κ, all
Local LLM, Qwen2.5 7B	0.70	0.45	0.41
FinBERT tone	0.65	0.31	0.20
NLI zero-shot (mDeBERTa)	0.63	0.29	0.18
SBERT classifier	0.61	0.21	0.17

Scores on the 100 texts written by economists; last column: κ on all 177 hand-coded texts. Thresholds and classifiers by 5-fold cross-validation.

Table 13. FinBERT tone against reading

FinBERT ↓ / reading →	negative	neutral	positive	Precision
negative	17	11	2	0.57
neutral	2	25	3	0.83
positive	1	10	19	0.63

90 sentences from economists' texts, 30 per FinBERT label, read blind to the label: agreement 68%, $\kappa = 0.52$.

Table 14. Stance by reading, texts written by economists

Actor	n	Favourable	Neutral	Critical
Research economists	50	8 [3–19]	90 [79–96]	2 [0–10]
Central banks and institutions	50	52 [39–65]	42 [29–56]	6 [2–16]

Percentages with Wilson 95% intervals.

D Additional tables and figures

Table 15 lists the policy events used in the event study and the test of tone (Sections 6 and 8), with the attention each one received. Table 16 lists every topic found by the topic model (Section 5).

Table 15. Policy events and the attention they received

Date	Event	Texts	Expected	Ratio	p
18 Jun 2019	Facebook announces Libra	1	3.4	0.3	0.83
02 Oct 2020	ECB digital euro report	13	4.5	2.9	0.01
20 Oct 2020	Bahamas launches Sand Dollar	17	5.7	3.0	0.01
14 Jul 2021	ECB investigation phase	6	13.0	0.5	0.93
25 Oct 2021	Nigeria launches eNaira	9	11.4	0.8	0.69
20 Jan 2022	Fed CBDC discussion paper	14	10.1	1.4	0.24
09 Mar 2022	US order on digital assets	6	10.3	0.6	0.84
01 Nov 2022	India e-rupee pilot	14	14.6	1.0	0.55
07 Feb 2023	UK digital pound consultation	10	13.8	0.7	0.76
28 Jun 2023	EU digital euro bill	32	19.3	1.7	0.05
18 Oct 2023	ECB preparation phase	63	19.1	3.3	0.00
23 Jan 2025	US order banning a US CBDC	26	21.9	1.2	0.29

Texts by economists dated to the day, in the 15 days from the event; expected from the daily rate over days −90 to −15. p: share of 500 random dates with a larger $z = (N - E)/\sqrt{E}$.

Table 16. Topics of the economists' corpus

	Topic (top terms)	Texts	2014–2019	2020–2022	2023–2026
1	cryptocurrencies, implementation, private, traditional, national	470	18.5	13.5	8.5
2	design, retail, legal, private, privacy	439	10.4	11.2	8.6
3	uncertainty, rates, design, rate, interest	349	14.2	8.7	6.7
4	dollar, trade, debt, reserve, inflation	326	5.7	5.8	7.5
5	deposits, interest, deposit, rate, effects	319	2.8	5.0	7.3
6	governor, innovation, fintech, kong, hong	306	17.5	11.2	4.7
7	russian, russia, ruble, federation, legal	238	0.0	5.1	4.9
8	china, rmb, e-cny, yuan, internationalization	213	1.4	5.3	4.6
9	cashless, evolution, riksbank, forms, physical	204	6.6	4.9	4.1
10	fintech, technologies, services, innovation, finance	199	7.1	2.9	4.4
11	euro, european, ecb, area, legal	186	0.5	3.6	4.2
12	cryptocurrencies, crypto, bitcoin, cryptocurrency, assets	186	9.0	4.0	3.6
13	journal, list, approved, chapter, implementation	176	0.5	3.4	4.2
14	india, rupee, indian, rbi, upi	132	0.0	1.1	3.5
15	ruble, russia, russian, form, platform	130	0.0	1.9	3.2
16	cross-border, interoperability, settlement, instant, mbridge	127	0.9	2.1	2.9
17	euro, european, europe, ecb, parliament	127	0.9	2.4	3.0
18	euro, sie, european, auswahl, textabschnitt	110	0.5	1.3	2.8
19	nigeria, enaira, african, africa, inclusion	103	0.5	1.6	2.4
20	legal, assets, virtual, law, regulation	101	0.9	1.5	2.3
21	euro, project, rulebook, latest, in-depth information	77	0.5	0.6	2.1
22	raqamli, markaziy, circular, tahlil, joriy	73	0.0	0.7	1.6
23	ukraine, e-hryvnia, ukrainian, electronic, civil	69	1.4	1.3	1.6
24	ruble, legal, civil, agreement, platform	50	0.0	0.6	1.3

BERTopic on 4,710 texts (multilingual SBERT embeddings, UMAP, HDBSCAN with leaf selection and at least 40 texts per topic); 50% of texts were outliers, assigned to the nearest topic. Terms: class-based TF-IDF on English texts. Period columns: % of the period's texts.

E Data and code availability

E.1 Data

The corpus is a single file, `data/cbdc_corpus.csv`: one row per document and 49 columns, in UTF-8. Every record keeps its issuer, its date with the precision of that date, its original language, its keywords (author, indexed and extracted), its abstract, introduction and conclusion when available, its full text, its relevance label and theme score, and the sources it was found in. The harvest ran from 8 to 11 October 2026; it covers 48,188 documents, of which 17,709 have CBDC as a theme and 4,710 were written by economists (Section B). The build step can also export it in two formats read by bibliometric software: a Scopus-style CSV (for `bibliometrix` and VOSviewer) and RIS.

Caches hold the results that are slow to recompute: sentence embeddings, sentence tones, the language model's stance labels, and the OpenAlex fields and institutions of each paper. The hand-coded validation samples and every coder's predictions are in `presentation/` (Table 9).

Many full texts are under copyright. The data are therefore shared with the members of the project and on request, not published openly. The metadata, identifiers (DOI, OpenAlex and source identifiers, URLs) and the code that rebuilds the corpus from the public sources are not restricted.

E.2 Code

All code is in Python and kept in a version-controlled repository, available from the authors on request. One command per step rebuilds the corpus, and every step resumes where it stopped:

1. `run.py harvest`: query every source (Section B);
2. `run.py fulltext`: download and extract the full texts;
3. `run.py build`: merge, de-duplicate, enrich and score relevance;
4. `scripts/tag_economists.py`: retrieve the OpenAlex field and institutions of each paper;
5. `scripts/validate_coders.py`: score every candidate coder against the hand-coded samples;
6. `scripts/make_appendix.py`: write the tables of this appendix (`paper/tables/`).

The analyses (networks, topics, motivations, tone, stance, event studies, robustness) are in `cbdc_biblio/analysis.py` and `cbdc_biblio/nlp.py`.

E.3 Software and models

Python 3.10 or later; the packages and their versions are listed in `requirements.txt`. The models are open and run locally:

- sentence embeddings: `paraphrase-multilingual-MiniLM-L12-v2` (Sentence-Transformers);
- tone: FinBERT (`ProsusAI/finbert`);
- natural language inference (tested coder): `mDeBERTa-v3-base-xnli-multilingual-nli-2mil7`;
- stance: Qwen2.5 7B Instruct, 4-bit, run with `mlx-lm` (Apple Silicon); Qwen2.5 3B was tested.

Everything ran on a laptop (Apple M1, 8 GB of memory). Coding new texts with the language model requires Apple Silicon; every other step runs on any machine once the caches are present.

F Use of artificial intelligence and statement of originality

F.1 Use of artificial intelligence

Artificial intelligence was used in two distinct ways in this work.

As research instruments. Several language models are part of the method and are described with it: multilingual sentence embeddings (topics), FinBERT (tone), a natural language inference model and two local Qwen2.5 models (tested as coders; the 7B model codes stance). Each of them was evaluated against a reading of the texts before being used, and their scores are reported in Section C. They run locally; no text of the corpus was sent to an external service for coding.

As assistants. A general-purpose AI assistant helped to write and debug the code that collects, cleans and analyses the corpus, to produce tables and figures, and to draft parts of the technical description in the appendices. The hand coding of the validation samples was AI-assisted: texts were read and coded blind to the machine labels, with the help of the assistant. For this reason, the validation scores measure agreement with an AI-assisted reading.

The research question, the choice and interpretation of the methods, the reading of the results and the conclusions are the authors'. Every number reported in the paper is computed by the code from the data, and the authors checked the text, the code and the results; they take full responsibility for the content of the paper.

F.2 Statement of originality

The authors declare that this paper is their own work. It does not reproduce, in whole or in part, text written by others without quotation marks and a reference; every idea, result, data source and software taken from other works is cited. The corpus consists of documents published by their authors; it is used for analysis only, and no full text from it is reproduced in the paper. The text produced with the help of an AI assistant was read, checked and edited by the authors, and was not copied from any other source.